

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE**Paper 2 (Generated practice paper - Set 04)**

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours**Materials**

For this paper you may use:

- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
 - use good English
 - organise information clearly
 - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
TOTAL	

Answer all questions in the spaces provided.

0 1 . 1

Which technique would be most suitable for detecting whether great crested newts are present in a pond without catching them? Tick **one** box.

[1 mark]

A Environmental DNA (eDNA) analysis

☐

B Radio tracking

☐

C Lincoln index (mark-release-recapture)

☐

D Leg rings

☐

0 1 . 2

Explain why geolocator tags, rather than radio or GPS transmitters, are used to track the migrations of small birds.

[2 marks]

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0 1 . 3

Explain how DNA databases could help to control the illegal trade in elephant ivory.

[2 marks]

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5

0 2 . 1

Explain **three** reasons why the development of new human activities has rarely been restricted by their environmental impacts.

[3 marks]

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0 2 . 2

Explain why sedentary rural communities often exploit resources sustainably.

[2 marks]

0 2 . 3

Explain why urban populations are often unaware that the resources they use are being depleted.

[2 marks]

0 2 . 4

Explain why the growth of affluence has increased per capita environmental impacts.

[3 marks]

10

03 . 1

Explain how natural material cycles differ from linear human systems of material use.

[4 marks]

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03 . 2

Explain why the use of toxic metals in electronic products is a particular problem for sustainability.

[2 marks]

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03 . 3

Describe how the design of a product can move it from a linear to a cyclical system.

[4 marks]

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0 4 . 1

Explain why a moderate water shortage reduces crop productivity even though the plants survive.

[3 marks]

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0 4 . 2

Explain the problems caused by waterlogged soils and describe **two** ways farmers reduce soil water levels.

[4 marks]

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0 4 . 3

Explain how irrigating with groundwater can cause soil salinisation and how this harms crops.

[2 marks]

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0 4 . 4

State how a farmer could raise the pH of an acidic soil and explain why a low pH reduces fertility.

[1 mark]

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10

0 5 . 1

Which feature of the Earth prevents most atmospheric gases from escaping into space? Tick **one** box.

[1 mark]

A Its mass

☐

B Its magnetic field

☐

C Its distance from the Sun

☐

D Its speed of rotation

☐

0 5 . 2

Explain how the speed of rotation of the Earth and the tilt of its axis affect the conditions for life.

[2 marks]

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0 5 . 3

Explain why the presence of liquid water on Earth depends on **both** the distance from the Sun **and** atmospheric pressure.

[2 marks]

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5

06

Table 1 shows the estimated population of blue whales and the number killed by whaling in selected years.

Table 1

Year	1900	1930	1960	1986	2020
Estimated population	240 000	100 000	5 000	2 000	15 000
Whales killed that year	200	30 000	1 200	0	0

06

1

Calculate the percentage decline in the blue whale population between 1900 and 1986.

[1 mark]

Answer

%

06

2

Explain why whale populations were so easily over-exploited.

[3 marks]

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06

3

Describe how the International Whaling Commission regulates whaling.

[4 marks]

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06 . 4

Explain why the ITTO has been less successful than the IWC in achieving sustainable exploitation.

[2 marks]

10

07

Table 1 shows the ecological footprint and biocapacity per person for four countries and the world in 2016.

Table 1

	Ecological footprint / gha per person	Biocapacity / gha per person	Human Development Index
USA	8.1	3.6	0.92
UK	4.4	1.4	0.92
China	3.6	1.0	0.75
Bangladesh	0.8	0.4	0.61
World	2.8	1.7	0.72

07 . 1

Define **ecological footprint** and **biocapacity**.

[2 marks]

07 . 2

Explain what is meant by a **global hectare** and why it is used.

[2 marks]

07 . 3

Use Table 1 to calculate how many Earths would be needed if everyone lived like the average US citizen.

[1 mark]

Answer

Earths

07 . 4

Use Table 1 to describe the relationship between development and ecological footprint, and explain the concept of Earth Overshoot Day.

[3 marks]

07 . 5

Suggest **two** ways a country such as the UK could reduce its footprint without reducing its HDI.

[2 marks]

1

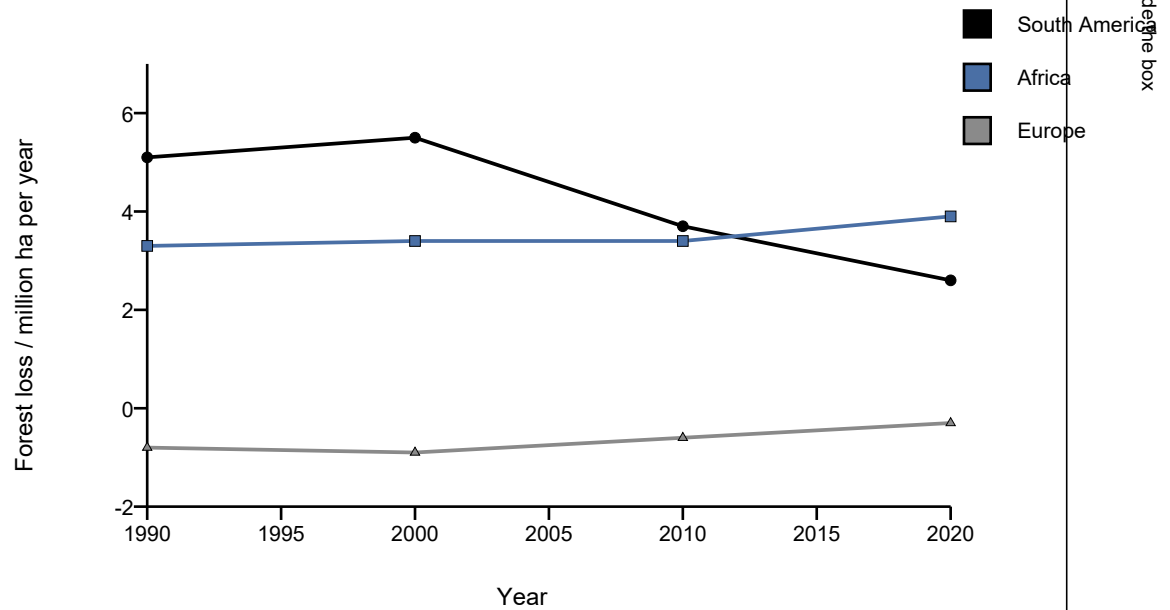
2

10

0 8

Figure 1 shows the area of forest lost each year in three regions between 1990 and 2020.

Figure 1



0 8

. 1

Describe the trends shown in Figure 1.

[3 marks]

0 8

. 2

Explain the main causes of forest loss in South America and Africa.

[4 marks]

08 . 3

Explain why roads and railways cause more forest loss than the area they occupy.

[2 marks]

08 . 4

Explain why softwood from Scandinavia and Canada is generally produced more sustainably than tropical hardwood.

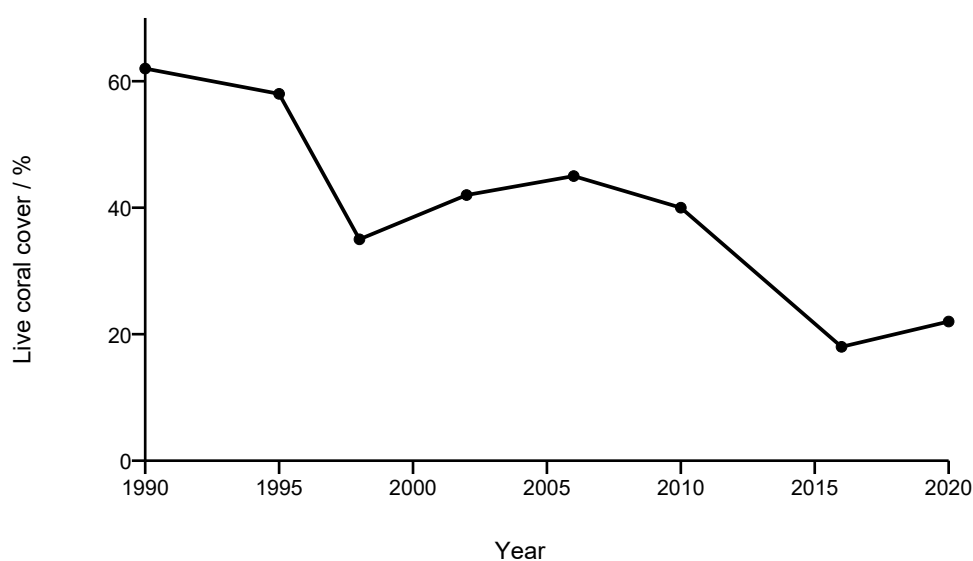
[1 mark]

10

09

Figure 1 shows the percentage of live coral cover on a reef near a growing coastal resort between 1990 and 2020, and the years of two major bleaching events.

Figure 1



09 . 1

Describe the changes in coral cover shown in Figure 1.

[2 marks]

09 . 2

Explain what coral bleaching is and give **three** causes.

[4 marks]

09 . 3

Explain how the growing resort could threaten the reef other than through bleaching.

[3 marks]

09 . 4

Explain why a no-take zone on the reef could increase fish catches in surrounding areas.

[1 mark]

10

1 0 . 1

Explain why the potential to increase food supplies by cultivating new land is limited.

[2 marks]

1 0 . 2

Explain why importing food may reduce the availability of food in the area where it is produced.

[2 marks]

1 0 . 3

Explain why organic agriculture is more than simply 'not using pesticides and artificial fertilisers'.

[2 marks]

1 0 . 4

Evaluate the strategies that could be used to make agriculture more sustainable.

[9 marks]

[illegible]

15

Write an essay on one of the following topics.

$$\boxed{1} \boxed{1} . \boxed{1}$$

Explain how the physical features of the Earth and the activities of living organisms have combined to create and maintain conditions suitable for life.

[25 marks]

OR

1 1 . 2

Discuss the extent to which the resources and services provided by other species justify the conservation of biodiversity.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS